PHYSICS

1)

Value of $(1 - \sin^2 37°) - (\cos^2 37° - 1)$ = ...

(1) 1
(2) - 1
(3) 2
(4) 3

2) Value of cos 74° is :-

(1) $\frac{24}{25}$
(2) $\frac{7}{25}$
(3) $-\frac{16}{25}$
(4) $-\frac{7}{25}$

3) A circular arc of length 15 cm has radius 5 cm. The angle subtended by it at centre is :-

(1) $\frac{1}{3}$ rad
(2) 3 rad
(3) 5 rad
(4) $\frac{1}{5}$ rad

4) y-intercept of straight line $\frac{x}{2} + \frac{y}{4} = 1$ is :-

(1) 2
(2) 4
(3) –2
(4) –4

5)

The value of cos α is $\frac{12}{13}$ find the value of sin2α :-

(1) $\frac{120}{169}$

(2) $\frac{169}{120}$

(3) $\frac{159}{120}$

(4) $\frac{223}{440}$

6) Find the value of $\sqrt{0.999}$:-

(1) 0.333

(2) 0.995

(3) 0.9995

(4) 0.9999

7) The least value of 4cos x + 3sin x is :-

(1) 5

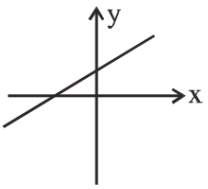
(2) –5

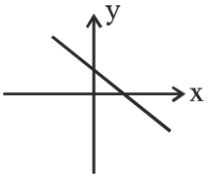
(3) 7

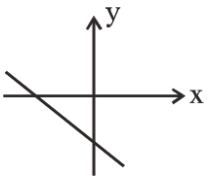
(4) –7

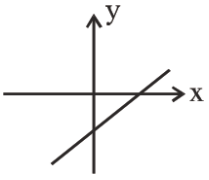
8)

Find out possible graph for equation
4x – 3y + 12 = 0

(1) 

(2) 

(3) 

(4) 

9) $\int_0^{\pi/2} \sin^2\theta \, d\theta$ =

(1) 1

(2) 0

(3) $\frac{\pi}{2}$

(4) $\frac{\pi}{4}$

10) Value of $\cos\left(12\pi + \frac{\pi}{6}\right)$ is :-

(1) $\frac{1}{2}$

(2) $-\frac{1}{2}$

(3) $\frac{\sqrt{3}}{2}$

(4) $\frac{-\sqrt{3}}{2}$

11) The value of the sum of the infinite series is $S = 1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \frac{1}{16} + ...\infty$

(1) 1

(2) $\frac{2}{3}$

(3) $\frac{3}{2}$

(4) 0

12) If α, β be the roots of the quadratic equation $x^2 - 10x + 24 = 0$ then $|\alpha - \beta|$ is :-

(1) –2

(2) 2

(3) 10

(4) 4

13) If $\tan\theta = \frac{1}{\sqrt{5}}$ and θ lies in the first quadrant, the value of cos θ is :

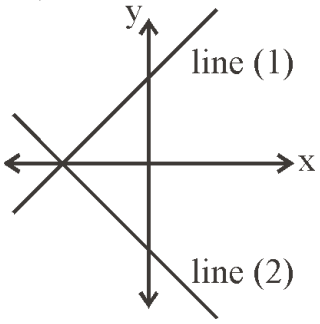
(1) $\sqrt{\frac{5}{6}}$

(2) $-\sqrt{\frac{5}{6}}$

(3) $\frac{1}{\sqrt{6}}$

(4) $-\frac{1}{\sqrt{6}}$

14) Which of the following statement is not correct for following straight line graph :-



(1) line (2) has negative y intercept

(2) line (1) has positive slope

(3) line (2) has negative slope

(4) line (1) has negative y intercept

15) Find $\int_{n-1}^{n} (3 + 4t)\, dt$:-

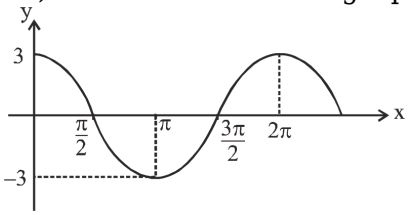
(1) 4n + 1

(2) 4n + 5

(3) $2n^2 + 1$

(4) $4n^2 + 5$

16) Which of the following equation is the best representation of the given graph ?



(1) y = 3 sin x

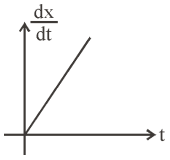
(2) y = sin 3 x

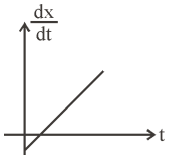
(3) y = 3 cos x

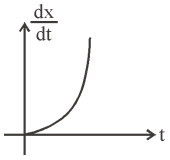
(4) y = cos 3 x

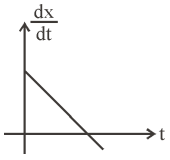
17) For function $x = t^2$, $\frac{dx}{dt}$ is shown by :-

(1)



(2) 

(3) 

(4) 

18) If $y = \sin(x^2 - 4)$, then $\frac{dy}{dx}$ is :-

(1) $x^2 \cdot \sin(x^2 - 4)$

(2) $2x \cdot \sin(x^2 - 4)$

(3) $x^2 \cdot \cos(x^2 - 4)$

(4) $2x \cdot \cos(x^2 - 4)$

19) The minimum value of $y = x^2 - 10x + 40$ is :-

(1) 2

(2) 5

(3) 15

(4) 24

20) If side length of cube is changing at 0.1 mm sec^{-1} . Then what will be the rate of change of volume, when side length is 2m.

(1) $8 \times 10^{-4}\ m^3\ sec^{-1}$

(2) $4 \times 10^{-3}\ m^3\ sec^{-1}$

(3) $12 \times 10^{-4}\ m^3\ sec^{-1}$

(4) $12 \times 10^{-3}\ m^3\ sec^{-1}$

21) $\int_R^\infty \frac{GMm}{x^2} dx =$

(1) $\frac{GMm}{R}$

(2) $-\frac{GMm}{R}$

(3) 0

(4) $\frac{GMm}{R^2}$

22) Area bounded by curve y= sinx, with x-axis when x varies from 0 to π/2 is.

(1) 1 Unit
(2) 2 Unit
(3) 3 Unit
(4) 0 Unit

23) A ball is projected with speed u at an angle θ to the horizontal. The range R of projectile is given by

$$R = \frac{u^2 \sin 2\theta}{g}$$

for which value of θ will the range be maximum (u and g are fixed)

(1) $\frac{\pi}{2}$radian
(2) $\frac{\pi}{4}$radian
(3) $\frac{\pi}{3}$radian
(4) $\frac{\pi}{6}$radian

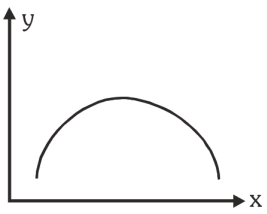
24) A particle moves along the curve $12y = x^3$. Which coordinate changes at faster rate at x = 10 :-

(1) X - coordinate
(2) Y - coordinate
(3) Both X and Y - coordinate
(4) Data insufficient

25) Which of the following graph is a straight line for equation $y^2 = 2x$?
(a) y v/s x^2 (b) y^2 v/s x
(c) y v/s $\sqrt{x}$ (d) $\sqrt{y}$ v/s x

(1) a and d
(2) b and c
(3) a and c
(4) only b

26) Magnitude of slope of the shown graph.



(1) First increases then decreases
(2) First decreases then increases
(3) Increases
(4) Decreases

27) If velocity of a particle is given by v = (2t + 3) m/s, then average velocity in interval $0 \le t \le 1$s is :

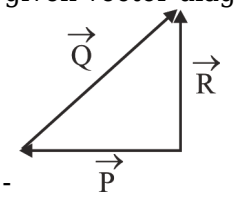
(1) $\frac{7}{2}$ m/s

(2) $\frac{9}{2}$ m/s

(3) 4 m/s

(4) 5 m/s

28) In the given vector diagram it is shown that $\vec{R} \perp \vec{P}$ and $|\vec{R}| = \frac{|\vec{Q}|}{2}$. The angle between the vectors $\vec{P}$ and $\vec{Q}$ is :-



(1) 30°

(2) 60°

(3) 120°

(4) 150°

29) Resultant of which of a following may be equal to zero ?

(1) 10N, 10N, 10N

(2) 10N, 10N, 25N

(3) 10N, 10N, 35N

(4) None of these

30) What is the angle between $\vec{A}$ and the resultant of $\left(\vec{A}+\hat{B}\right)$ and $\left(\vec{A}-\hat{B}\right)$?

(1) 0°

(2) $\tan^{-1}\left(\frac{A}{B}\right)$

(3) $\tan^{-1}\left(\frac{B}{A}\right)$

(4) $\tan^{-1}\left(\frac{A-B}{A+B}\right)$

31) A unit vector in the direction of resultant vector of $\vec{A} = -2\hat{i} + 3\hat{j} + \hat{k}$ and $\vec{B} = \hat{i} + 2\hat{j} - 4\hat{k}$ is :-

(1) $\frac{-2\hat{i} + 3\hat{j} + \hat{k}}{\sqrt{35}}$

(2) $\frac{-\hat{i}+2\hat{j}+4\hat{k}}{\sqrt{35}}$

(3) $\frac{-\hat{i}+5\hat{j}-3\hat{k}}{\sqrt{35}}$

(4) $\frac{-3\hat{i}+\hat{j}-5\hat{k}}{\sqrt{35}}$

32) If $\frac{\left|\vec{a}+\vec{b}\right|}{\left|\vec{a}-\vec{b}\right|}$ = 1, then angle between $\vec{a}$ and $\vec{b}$ is :-

(1) 0°

(2) 45°

(3) 90°

(4) 60°

33) Five equal forces of 10N each are applied at one point and all are lying in one plane. If the angles between them are equal, the resultant force will be:

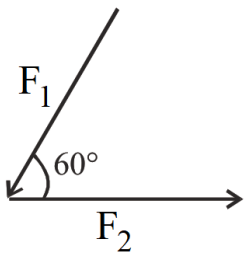
(1) Zero

(2) 10N

(3) 20N

(4) $10\sqrt{2}$N

34) Two forces, each equal to 7N, act as shown in fig. Their resultant is :-



(1) 7/2

(2) 7

(3) $7\sqrt{3}$

(4) $7\sqrt{2}$

35) The resultant of two forces 3P and 2P is R. If the first force is doubled then the resultant is also doubled the angle between the two force is :-

(1) 60°

(2) 90°

(3) 120°

(4) 180°

36) The position vectors of two balls are given by

$\vec{r}_1 = 2(m)\,\hat{i} + 7(m)\hat{j}$
$\vec{r}_2 = -2(m)\,\hat{i} + 4(m)\hat{j}$
What will be the distance between the two balls?

(1) 7 m
(2) 5 m
(3) 4 m
(4) 3 m

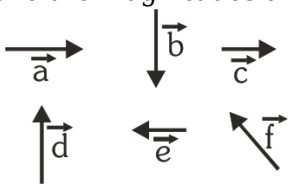
37) The magnitudes of vectors $\vec{A}$, $\vec{B}$ and $\vec{C}$ are respectively 7, 24 and 25 units and $\vec{A} + \vec{B} = \vec{C}$, then the angle between $\vec{A}$ and $\vec{B}$ is :

(1) 0
(2) π/3
(3) π/2
(4) π/4

38) If vectors $\vec{A}$ and $\vec{B}$ are such that $\left|\vec{A} + \vec{B}\right| = \left|\vec{A}\right| = \left|\vec{B}\right|$, then $\left|\vec{A} - \vec{B}\right|$ may be equated to

(1) $\frac{\sqrt{3}}{2}\left|\vec{A}\right|$
(2) $\left|\vec{A}\right|$
(3) $\sqrt{2}\left|\vec{A}\right|$
(4) $\sqrt{3}\left|\vec{A}\right|$

39) Six vectors, $\vec{a}$ through $\vec{f}$ have the magnitudes and directions indicated in the figure. Which of the following statements is true ?



(1) $\vec{b} + \vec{e} = \vec{f}$
(2) $\vec{b} + \vec{c} = \vec{f}$
(3) $\vec{d} + \vec{c} = \vec{f}$
(4) $\vec{d} + \vec{e} = \vec{f}$

40) At what angle must the two forces (x + y) and (x – y) act so that the resultant may be $\sqrt{(x^2 + y^2)}$?

(1) $\cos^{-1}\left[\frac{-(x^2 + y^2)}{2(x^2 - y^2)}\right]$

(2) $\cos^{-1}\left[\frac{-2(x^2-y^2)}{x^2+y^2}\right]$

(3) $\cos^{-1}\left[\frac{-(x^2+y^2)}{x^2-y^2}\right]$

(4) $\cos^{-1}\left[\frac{(x^2-y^2)}{x^2+y^2}\right]$

41) Vector sum of two forces of 10N and 6N cannot be :

(1) 4N

(2) 8N

(3) 12N

(4) 2N

42)

List-I		List-II	
(I)	$\frac{d}{dx}(x\ell nx)$	(P)	$\frac{1}{(2x+1)^2}$
(II)	$\frac{d}{dx}\left(\frac{x}{2x+1}\right)$	(Q)	12
(III)	$\frac{d^2}{dx^2}\left(6x^2\right)$	(R)	$1+\square nx$
(IV)	$\frac{d}{dx}\left(\frac{1}{2x+1}\right)$	(S)	$-\frac{2}{(2x+1)^2}$

(1) I → R; II → P; III → Q; IV → S

(2) I → S; II → R; III → Q; IV → P

(3) I → P; II → Q; III → R; IV → S

(4) I → Q; II → S; III → P; IV → R

43) The velocity of a particle is given by the equation $v = \sqrt{\sin 2t}$. The value of acceleration of particle at $t = \frac{\pi}{4}\sec$.

(1) $\frac{1}{\sqrt{2}}$

(2) 1

(3) 0

(4) ∞

44) $\frac{d}{dx}\left(3x^3+4\sin x-3\cos x\right)$ is equal to

(1) $9x^2 - 4\cos x + 3\sin x$

(2) $9x^2 + 4\cos x - 3\sin x$

(3) $9x^2 + 4\cos x + 3\sin x$

(4) $9x^2 - 4\cos x - 3\sin x$

45)

If x and y are function of another variable t as

$x = 3t^2$, $y = 2t^3$, then the value of $\frac{dy}{dx}$ is

(1) t

(2) 2t

(3) $\frac{t}{2}$

(4) None

CHEMISTRY

1) Which experiment led to the discovery of the nucleus?

(1) Cathode ray experiment

(2) Gold foil experiment

(3) Oil drop experiment

(4) Photoelectric effect

2) The increasing order for the values of *e/m* (charge/mass) is :

(1) e, p, n, α

(2) n, p, e, α

(3) n, p, α, e

(4) n, α, p, e

3) The energy ratio of a photon of wavelength 3000Å and 6000Å is :-

(1) 1 : 1

(2) 1 : 4

(3) 2 : 1

(4) 1 : 2

4) $^{238}_{92}U$ is isodiapher with $^{234}_{Z}X$. The value of Z is.

(1) 90

(2) 88

(3) 92

(4) 86

5) Which one of the following sets of ions represents a collection of isoelectronic species ?

(1) Li^+, Na^+, Mg^{2+}, Ca^{2+}

(2) K^+, Cl^-, Ca^{2+}, S^{2-}

(3) Be^{2+}, Mg^{2+}, K^+, Ca^{2+}

(4) N^{3-}, O^{2-}, F^-, S^{2-}

6) If mass of electron become half, mass of proton become double and mass of neutron become $3/4^{th}$ than new mass of ${}_8O^{16}$ will be :-

(1) 37.5 % increase

(2) Remain unchange

(3) 12.5 % increase

(4) 25 % decrease

7) Atomic number and mass number of an element M are 25 and 52 respectively. The number of electrons, protons and neutrons in M^{+2} ions are respectively :-

(1) 25, 25 and 27

(2) 25, 27 and 25

(3) 27, 25 and 27

(4) 23, 25 and 27

8) The number of photons in a radiation of wavelength 662.6 nm that provide 3 J of energy are :

(1) 10^9

(2) 10^{19}

(3) 10^{28}

(4) 0.33×10^9

9) A dye absorbs a photon of wavelength λ and re-emits the same energy into two photons of wavelength λ_1 & λ_2 respectively. The wavelength is related with λ_1 & λ_2 as

(1) $\lambda = \dfrac{\lambda_1 + \lambda_2}{\lambda_1 \lambda_2}$

(2) $\lambda = \dfrac{\lambda_1 \lambda_2}{\lambda_1 + \lambda_2}$

(3) $\lambda = \dfrac{\lambda_1^2 \lambda_2^2}{\lambda_1 + \lambda_2}$

(4) $\lambda = \dfrac{\lambda_1 \lambda_2}{(\lambda_1 + \lambda_2)^2}$

10) An ion has a charge of –3. It has 7 protons. No. of electrons in ion is :-

(1) 7

(2) 10

(3) 4

(4) 14

11) When an electron come close to the nucleus :

(1) its kinetic energy increases, potential energy and total energy decrease

(2) kinetic energy and total energy increase, potential energy decreases

(3) kinetic energy and total energy decrease, potential energy increases

(4) kinetic energy, potential energy and total energy increase.

12) **Assertion (A) :** During Rutherford's scattering experiment few positively charged α-particles were deflected.
Reason (R) : The positive charge of the atom is spread throughout the atom.

(1) Both A and R are true and R is the correct explanation of A.

(2) Both A and R are true but R is not the correct explanation of A.

(3) A is true but R is false.

(4) Both A and R are false.

13) Average atomic weight of an element M is 51.7. If two isotopes of M, M^{50} and M^{52} are present then calculate percentage occurance of M^{50} in nature.

(1) 85%

(2) 15%

(3) 75%

(4) 25%

14)

Maximum frequency will be of

(1) UV-rays

(2) IR

(3) Radiowaves

(4) X-rays

15)

What is the corresponding wave number of $\lambda = 5nm$?

(1) $2 \times 10^{-6} m^{-1}$

(2) $2 \times 10^{6} m^{-1}$

(3) $2 \times 10^{8} m^{-1}$

(4) $2 \times 10^{-9} m^{-1}$

16) Which of the following is not correct for electromagnetic waves?

(1) The wavelength is the distance between two successive crests
(2) The frequency is the number of waves which pass a given point in one second
(3) The velocity of a wave is the distance covered by the particular wave in one second
(4) All electromagnetic waves have equal wavelengths

17) Calculate the frequency of radiation having wavelength 1 Å ?

(1) 3×10^{18} Hz
(2) 3×10^{18} sec^{-1}
(3) 3×10^{18} cycles/sec.
(4) All of the above

18) The ratio of first three radius ratio of Bohr orbit of H atom is –

(1) 1 : 0.5 : 0.5
(2) 1 : 2 : 3
(3) 1 : 8 : 27
(4) 1 : 4 : 9

19) The ratio of minimum and maximum wavelengths of Lyman series in spectrum of Li^{+2} ion is

(1) 4:27
(2) 3:4
(3) 4:2
(4) 4:3

20) The maximum number of electrons present in a subshell having = 3, is :

(1) 6
(2) 8
(3) 10
(4) 14

21) In Mn how many electron having n = 2, = 0 ?

(1) 2
(2) 8
(3) 25
(4) 1

22) For how many orbitals n + = 6 and m = 0 is possible

(1) 2
(2) 3
(3) 4

(4) 5

23) Which of the following statement is true :-
(a) 6 electrons are present in Mg for which m = 0
(b) Maximum 6 electrons can accomdate in p_x-orbital
(c) Maximum 18 electrons present in M-shell
(d) 3–electron present in phosphorous for which,
$\ell = 0$, $S = +\frac{1}{2}$

(1) a, b, c, d
(2) a, b c
(3) a, c, d
(4) c, d

24) If the speed of electron in first Bohr orbit of hydrogen atom be x, then speed of the electron in 3rd orbit is

(1) x/9
(2) x/3
(3) 3x
(4) 9x

25) The radius of the nucleus is related to the mass number A by :-

(1) $R = R_0 (A)^{1/3}$
(2) $R = R_0 (A)$
(3) $R = R_0 (A)^{1/2}$
(4) $R = R_0 (A)^{1/4}$

26) If the table of atomic masses were established with the oxygen atom and assigned value of 100, then the mass of carbon atom would be -

(1) 24
(2) 75
(3) 50
(4) 112

27) Which of the following quantum number set is not possible :-

(1) n = 3, ℓ = 2, m = +1, S = –1/2
(2) n = 4, ℓ = 1, m = +2, S = +1/2
(3) n = 4, ℓ = 3, m = 0, S = +1/2
(4) n = 3, ℓ = 1, m = 0, S = –1/2

28) Number of de-Broglie waves in sixth excited state of hydrogen atom is :-

(1) 7
(2) 6
(3) 5
(4) 8

29) If total energy of electron in H-atom is –1.51 eV. Then potential energy will be :-

(1) –2.02 eV
(2) –3.02 eV
(3) –1.51 eV
(4) +3.02 eV

30) The energy of electron in H-atom in ground state is –13.6 eV. The energy in n = 4 will be :-

(1) –0.54 eV
(2) –0.85 eV
(3) –3.4 eV
(4) –1.51 eV

31) The highest velocity of electron will be for :-

(1) Li^{+2}, n = 3
(2) Be^{+3}, n = 6
(3) He^{+}, n = 8
(4) H, n = 4

32) Angular momentum of M-shell electron is :-

(1) $\frac{h}{2\pi}$
(2) 0
(3) $\frac{3h}{\pi}$
(4) $\frac{3h}{2\pi}$

33) 4d, 5p, 5f and 6p-orbitals are arranged in the order of decreasing energy.
The correct option is :-

(1) 6p > 5f > 5p > 4d
(2) 6p > 5f > 4d > 5p
(3) 5f > 6p > 4d > 5p
(4) 5f > 6p > 5p > 4d

34) The number of radial nodes for 3 p-orbital is .....

(1) 3

(2) 4

(3) 2

(4) 1

35) Identify the correct increasing order of energy of the orbitals for hydrogen atom :

(1) 1s < 2s = 2p < 3s = 3p = 3d < 4s = 4p = 4d = 4f

(2) 1s > 2s = 2p > 3s = 3p = 3d > 4s = 4p = 4d = 4f

(3) 1s = 2s = 3s = 4s > 2p = 3p = 4p > 3d = 4d > 4f

(4) 1s = 2s = 3s = 4s < 2p = 3p = 4p < 3d = 4d < 4f

36) A 0.66 kg ball is moving with a speed of 100 m/s. The associated de broglie wavelength will be – ($h = 6.6 \times 10^{-34}$ J.S)

(1) 6.6×10^{-34} m

(2) 1×10^{-32} m

(3) 6.6×10^{-32} m

(4) 1×10^{-35} m

37) The value of principal quantum number is 3. The number of its (i) subshell and (ii) orbitals would be respectively.

(1) 3 and 5

(2) 3 and 7

(3) 3 and 9

(4) 2 and 5

38) The uncertainty in the position of a moving bullet of mass 10 gm is 10^{-5} m. Calculate the uncertainty in its velocity :-

(1) 5.2×10^{-28} m/sec

(2) 3×10^{-28} m/sec

(3) 5.2×10^{-22} m/sec

(4) 3×10^{-22} m/sec

39) Energy electron in third excited state for Be^{+3} is :-

(1) –13.6 eV

(2) –1.51 eV

(3) –0.85 eV

(4) –0.54 eV

40) Number of unpaired electrons in Cu^{2+} is :-

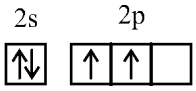
(1) Zero

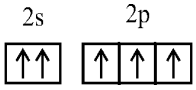
(2) 9

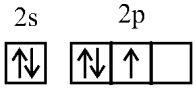
(3) 3

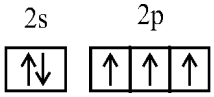
(4) 1

41) In which of the following electron distribution in ground state, only Hund's rule is violated?

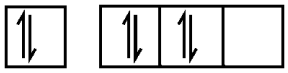
(1) 

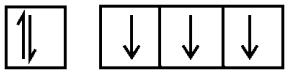
(2) 

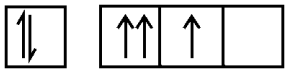
(3) 

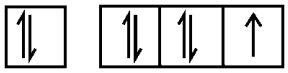
(4) 

42) The orbital diagram in which both the Pauli's exclusion principle and Hund's rule are violated is:-

(1) 

(2) 

(3) 

(4) 

43)

Column-I		Column-II	
(i)	Aufbau principle	(a)	$\lambda = h/(mv)$
(ii)	De Broglie's hypothesis	(b)	$\lambda = hc/\Delta E$
(iii)	Hund's rule	(c)	Max. unpaired electrons
(iv)	Planck's Theory	(d)	Electronic configuration

Correct match is :-

(1) (i) d, (ii) b, (iii) c, (iv) a

(2) (i) c, (ii) a, (iii) d, (iv) b

(3) (i) d, (ii) a, (iii) c, (iv) b

(4) (i) c, (ii) b, (iii) d, (iv) a

44) An electron in H atom jumps from the third energy level to the first energy level. The change in the total energy of the electron is :-

(1) 12.09 eV

(2) 6.04 eV

(3) 24.18 eV

(4) 3.01 eV

45) What is the ratio of energies of Li^{+2} for 2nd & I^{st} excited state :-

(1) 9 : 4

(2) 4 : 9

(3) 2 : 1

(4) 1 : 2

BIOLOGY

1)

Which of the following statement is correct regarding compound epithelum ?

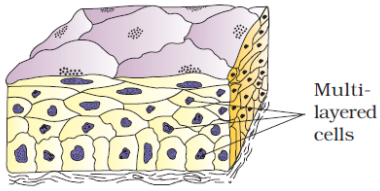
(1) Major function in secretion and absorption

(2) Limited role in protection

(3) It covers dry surface of skin, the moist surface of buccal cavity

(4) It is composed of single layer of cells

2)



Identify the option which includes only correct statements regarding the epithelium shown in figure.
(a) Made of more than one layer of cells.
(b) It is a simple epithelium.
(c) It has limited role in secretion and absorption.
(d) It's main function is to provide protection against chemical and mechanical stresses.
Option :

(1) (a), (c), and (d) only

(2) (a), (b) and (c) only

(3) (a) and (d) only

(4) (b), (c) and (d) only

3) Mark the incorrect match

	Column I	**Column II**
(1)	Strongest cartilage	White fibrous cartilage
(2)	Adipose tissue	Act as shock absorber

(3)	Spongy connective tissue	Areola
(4)	Bones	Hard and pliable matrix rich in Ca^{++} salt

(1) 1

(2) 2

(3) 3

(4) 4

4) Given below are two statement:
Statement-I: Bone marrow of all the bones is the site of production of blood cells
Statement-II: Cartilage resists compression
In the light of the above statements. Choose the most appropriate answer form the options given below.

(1) Both statement I and statement II are incorrect

(2) Statement I is correct and statement II is incorrect

(3) Statement I is incorrect and statement II is correct

(4) Both statement I and statement II are correct

5) Match the column:

	Tissue		**Function**
(a)	Cuboidal epithelium	(i)	To move mucus or particles in a specific direction
(b)	Compound	(ii)	Protection against mechanical stress
(c)	Ciliated epithelium	(iii)	Secretion of earwax
(d)	Glandular epithelium	(iv)	Secretion and absorption

(1) (a)-iii, (b)-ii, (c) -i, (d)-iv

(2) (a)-iv, (b)-i, (c) -iii, (d)-ii

(3) (a)-ii, (b)-iii, (c) -i, (d)-iv

(4) (a)-iv, (b)-ii, (c) -i, (d)-iii

6) **Assertion (A):** In frogs, cutaneous respiration can maintain life during hibernation and aestivation.
Reason (R): Frogs possess a three-chambered heart which pumps oxygenated and deoxygenated blood separately without any mixing.

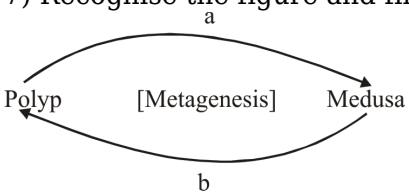
(1) Both A and R are true, and R is the correct explanation of A

(2) Both A and R are true, but R is not the correct explanation of A

(3) A is true, but R is false

(4) A is false, but R is true

7) Recognise the figure and find out the correct matching for 'a' and 'b' :-



(1) a-sexually, b-asexually

(2) a-asexually, b-sexually

(3) a-asexually, b-parthenogenetically

(4) a-sexually, b-parthenogenetically

8)

Identify the correct match from column I, II & III.

	Column-I		**Column-II**		**Column-III**
(1)	Taenia solium	(a)	Asymmetrical	(i)	Dimorphic
(2)	Sponge	(b)	Bilateral	(ii)	Exclusively marine habitat
(3)	Obelia	(c)	Radial	(iii)	Pseudo segmentation
(4)	Pleurobranchia	(d)	Ciliated comb plates	(iv)	Canal system

(1) 4-c-ii, 3-d-i, 2-a-iii, 1-b-iv

(2) 3-c-iii, 2-a-ii, 1-b-i, 4-d-iv

(3) 1-b-iii, 2-a-iv, 3-c-i, 4-d-ii

(4) 2-a-iv, 1-b-iii, 4-c-i, 3-d-ii

9) Read the following statements about the phylum porifera :-
(i) Commonly known as sponges
(ii) Are generally found in fresh water and mostly symmetrical animals
(iii) Possess water transport system
(iv) The body is supported by skeleton made up of spicules or spongin fibres
(v) Fertilization is external
Which of the above statements are incorrect?

(1) (iii), (v)

(2) (i), (ii), (iii)

(3) (i), (ii), (iii),(iv) and (v)

(4) (ii), (v)

10) Which of the following statement is incorrect for phylum cnidaria ?

(1) Possess cnidocytes which contain stinging capsule nematoblast.

(2) Stinging cells are used for anchorage defence and capture of prey.

(3) Digestion is extracellular and intracellular

(4) Exhibit two basic forms called polyp and medusa.

11) Which of the following option is correct and without exception ?

(1) Fertilization is internal and development is indirect- *Scypha*, *Ctenoplana*

(2) Flame cells help in osmoregulation and alimentary canal present-*Fasciola, Taenia*

(3) Body is segmented and performs locomotion with the help of parapodia- *Nereis, Pheretima*

(4) Body is covered by mantle layer and gills are present - *Pinctada, Pila*

12)

Read the following statements (A-D) and choose the correct option :-
(A) All animals are eukaryotes.
(B) Enterocoelomate animals are deutrostomiate.
(C) All triploblastic animals have coelom.

(D) All schizocoelomate animals have segmented body.

(1) A, B and D correct

(2) A and B incorrect

(3) Only A correct

(4) Only C and D incorrect

13) Simple squamous epithelium is present in :-
(A) Bowman's capsule
(B) Alveoli of lungs
(C) DCT of Nephron
(D) Fallopian tube
(E) Mesothelium

(1) A, B, C and E

(2) A, B, D and E

(3) A, B and E

(4) A, B and D

14) Path of water through a sponge is:

(1) Ostia → Spongocoel → Osculum

(2) Osculum → Spongocoel → Ostia

(3) Ostia → Spongocoel → Ostia

(4) Spongocoel → Ostia → Osculum

15) Parasitic adaptation of flatworm is:

(1) Presence of hooks and suckers

(2) Absorption of food through body surface
(3) Both (1) and (2)
(4) Presence of complex digestive system

16) What is the function of adhering junction?

(1) Cementing to keep neighbouring cells together.
(2) Give rigidity to tissue
(3) Stop leaking substance across the tissue
(4) All of these

17) Which cartilage is present at the end of long bones?

(1) Calcified cartilage
(2) Hyaline cartilage
(3) Elastic cartilage
(4) Fibrous cartilage

18) Translucent and bluish-white cartilage is

(1) Hyaline cartilage
(2) Elastic cartilage
(3) Fibrous cartilage
(4) Auricular cartilage

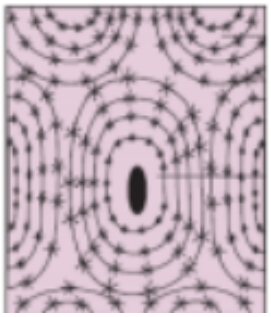
19) Ligaments connect

(1) Bones with muscles
(2) Bones with bones
(3) Muscles with muscles
(4) Skin with muscles

20) **Statement - I:** Macrophages act as internal scavengers
Statement - II: They engulf cell debris

(1) Both Statement I and II are correct
(2) Statement I is correct statement II is incorrect
(3) Statement I is incorrect and statement II is correct
(4) Both statement I and II are incorrect

21) Identify the given below tissue with its type and select the right option for the two together

 Option :

	Tissue	Type
(1)	Cartilage	Specialised connective tissue
(2)	Tendon	Dense irregular connective tissue
(3)	Ligament	Dense regular connective tissue
(4)	Bone	Specialised connective tissue

(1) 1
(2) 2
(3) 3
(4) 4

22) The major inorganic component of vertebrate bone is :-

(1) Sodium chloride
(2) Calcium carbonate
(3) Calcium Phosphate
(4) Magnesium Phosphate

23) In dense Connective tissue

(1) Matrix is more and fibres are less
(2) Both matrix and fibres are less
(3) Matrix is less and fibres are more
(4) Both matrix and fibres are more

24) In human fibrous cartilage is found abundantly

(1) Hyaline cartilage of joints
(2) Nostrils
(3) Intervertebral discs
(4) External ear

25) Which of the following statement is incorrect?

(1) The intercellular material of cartilage is solid and pilable and resists compression

(2) All the cartilages in vertebrae embryo are replaced by bones in adults

(3) Cartilage is present in the tip of nose, outer ear joints, between adjacent bones of vertebral column, limbs of adults

(4) Cells of cartilage are enclosed in small cavities within the matrix secreted by them

26) Mammalian pinna is supported by

(1) Hyaline cartilage

(2) Calcified cartilage

(3) Elastic cartilage

(4) White fibrous connective tissue

27) Match the following organism's with their respective characteristics -

a.	*Fasciola*	i	Parapodia
b	*Nereis*	ii	Flame cells
c	*Hydra*	iii	Bioluminescence
d	Comb jelly	iv	Cnidoblast cells

Select the correct option from the following -

(1) a. - (i), b - (ii), c - (iv), d - (iii)

(2) a. - (ii), b - (i), c - (iv), d - (iii)

(3) a. - (ii), b - (i), c - (iii), d - (iv)

(4) a. - (iii), b - (i), c - (iv), d - (ii)

28) Out of the given options choose the character present only in the coelenterates :-

(1) Hermaphroditism

(2) Flame cells

(3) Polymorphism

(4) Nematocysts

29) Phylum mollusca can be distinguised from other invertebrates by the presence of :

(1) Bilateral symmetry

(2) Coelom

(3) Mantle

(4) Gills

30) Match the following list of animals with their level of organisation :-

	Column-I		**Column-II**
(A)	Organ level	(i)	Pheretima

(B)	Cellular level	(ii)	Fasciola
(C)	Tissue level	(iii)	Spongilla
(D)	Organ system level	(iv)	Obelia

Choose the correct match showing division of labour with animal example :-

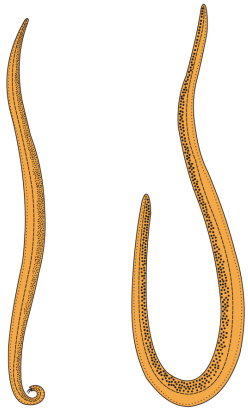
(1) A-iv, B-i, C-ii, D-iii

(2) A-iv, B-i, C-iii, D-ii

(3) A-ii, B-iii, C-iv, D-i

(4) A-i, B-iv, C-iii, D-ii

31) Which one of the following option are correct for members of phylum associated with the given figure:-



(1) Sexes are not separate.

(2) They have organ level of body organisation and acoelomate animals.

(3) Alimentary canal is complete with a well- developed muscular pharynx.

(4) Fertilisation is external and development is direct

32) Read the following match about phylum and its one character :
I. Hemichordata : Small worm–like marine animals with notochord.
II. Arthropoda : Chitinous exoskeleton with jointed appendages.
III. Mollusca : Feather like gills for excretion & respiration
IV. Ctenophora : Bioluminescence is well-marked.
Which one of the following from the above is/are the incorrect match?

(1) I, II

(2) II, III

(3) II, III, IV

(4) Only I

33) Consider the following set of animals and choose the option in which all animals belonging to the same phylum.

(1) *Sycon, Euspongia, Spongilla*

(2) *Aurelia, Obelia, Planaria*

(3) *Nereis, Pheretima, Locusta*

(4) *Asterias, Ophiura, Limulus*

34) Segments with a serial repetition of atleast some organs. This phenomena is called :-

(1) Metamorphosis
(2) Metamerism
(3) Metagenesis
(4) Metachrosis

35) Male frog can be distinguished from female frog by :–

(1) Anal style
(2) Truncus arteriosus
(3) Amplexusory pad and vocal sacs
(4) Teeth on lower jaw

36) Number of cranial nerves in frog are :-

(1) 8
(2) 10
(3) 12
(4) 20

37) Truncus arteriosus is present on _____ side of heart of frog :-

(1) Dorsal
(2) Lateral
(3) Ventral
(4) Dorso-lateral

38) Find out correct statement w.r.t to Rana tigrina:

(1) Frogs are beneficial to mankind because they eat insect and crop both
(2) Frogs are beneficial to mankind because they eat crop and protect insects
(3) Frog are beneficial to mankind because they eat insects and protect crops
(4) Frog are beneficial to mankind because they protect both insect and crops

39) Frog has no neck. Suppose it had one, it would

(1) breath easily by floating on water surface.
(2) Swim easily in water
(3) be unable to leap on land easily
(4) swallow its prey easily

40) Which of the following statements is incorrect about the frog ?
(i) Eyes are bulged out and covered by a nictitating membrane that protects them while in water.

(ii) On either side of the eyes a membranous tympanum (ear) receives sound signals.
(iii) The hind limbs end in four digits and they are larger and muscular than fore limbs that end in five digits.
(iv) Frogs are cold blooded animals.
(v) Frog exhibit sexual dimorphism

(1) (i) and (v)

(2) (iii) only

(3) (ii) and (iii)

(4) (iv) only

41) Frog is :-
(a) Ureotelic animal.
(b) Sexual dimorphic animal.
(c) Insectivorous animal.
(d) Cold blooded animal.
Which are correct about the frog?

(1) a and d

(2) b and c

(3) a, b and d

(4) a, b, c and d

42)

Match the following regarding frog :-

(A)	Nictitating membrane	i	large muscular with five digits
(B)	Tympanum	ii	receives sound signals
(C)	Hind limbs	iii	four digits
(D)	Fore limbs	iv	protects the eye when in water

(1) A - ii, B - iv, C - i, D - iii

(2) A - iv, B - ii, C - i, D - iii

(3) A - iv, B - ii, C - iii, D - i

(4) A - iv, B - i, C - ii, D - iii

43) **Assertion :** Frogs exhibits sexual dimorphism.
Reason : Male frogs can be distinguished by the presence of sound producing vocal sacs and also a copulatory pad on the first digit of the fore limbs which are absent in female frogs.

(1) Both Assertion and Reason are True and the Reason is a correct explanation of the Assertion.

(2) Both Assertion and Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion and Reason are False.

44) Match the column-I with column-II

Column-I		Column-II	
(A)	Hibernation	(i)	Winter sleep
(B)	Camouflage	(ii)	The ability to change the colour to hide them from their enemies
(C)	Poikilotherm	(iii)	Cold blooded
(D)	Aestivation	(iv)	Summer sleep

(1) A–iv, B–ii, C–iii, D-i

(2) A–i, B–ii, C–iv, D-iii

(3) A–i, B–ii, C–iii, D-iv

(4) A–i, B–iii, C–iv, D-ii

45)

Read the following statement carefully about frog :-
(A) The skin is smooth and slippery due to presence of mucus.
(B) Body is divisible into head, trunk and tail.
(C) Copulatory pad on the first digit of the hind limbs are present only in male frog.
(D) Frogs do not exhibit sexual dimorphism.
(E) The Indian bull frog, *Rana tigrina* is the common frog found in India.

Find out correct option :-

(1) only A, C, D are correct statements

(2) only B, C, D are incorrect statements

(3) only A, E are incorrect statements

(4) All are correct statements

46) Right auricle of frog contains ___(A)___ blood and it receives the blood from ___(B)___. Identify A and B :

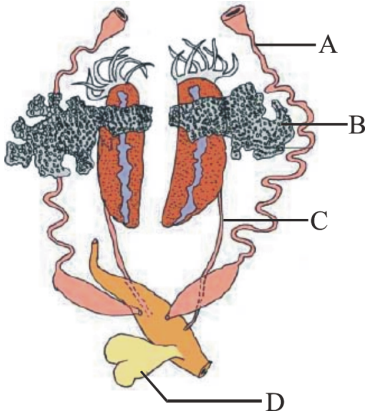
(1) A – Oxygenated, B – Sinus venosus

(2) A – Deoxygenated, B – Conus arteriosus

(3) A – Oxygenated, B – Truncus arteriosus

(4) A – Deoxygenated, B – Sinus venosus

47) Given below is a female reproductive system of frog labelled as A, B, C and D



Find out the incorrect one :-

(1) In male frog labelled C act as urinogenital duct

(2) In female frog labelled C and A open separately in the cloaca

(3) Labelled D is urinary bladder which is present ventral to rectum

(4) Labelled B is functionally connected to kidney.

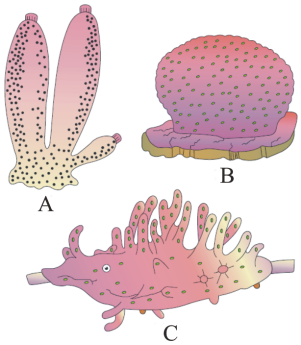
48) Brain coral is :-

(1) *Gorgonia*

(2) *Metridium*

(3) *Meandrina*

(4) *Adamsia*



49) Identify the figure A, B, C :- **Option :-**

	A	B	C
(1)	Euspongia	Spongilla	Sycon
(2)	Spongilla	Euspongia	Sycon
(3)	Spongilla	Sycon	Euspongia
(4)	Sycon	Euspongia	Spongilla

(1) 1

(2) 2

(3) 3

(4) 4

50) Sexes are seperate in _____ :-

(1) Aschelminthes

(2) Porifera

(3) Ctenophora

(4) Platyhelminthes

51) In some animal groups, the body is found divided into compartments with serial repetition of at least some organs. This is found in-

(1) Earthworm, Pila

(2) Earthworm, Rat

(3) Octopus, Starfish

(4) Balanoglossus, Cockroach

52)

	Column I		Column II
(a)	Arthropoda	(i)	Jointed appendages
(b)	Mollusca	(ii)	Radula
(c)	Echinodermata	(iii)	Proboscis and collar
(d)	Hemichordata	(iv)	Water vascular system

(1) a–i, b–ii, c–iii, d–iv

(2) a–i, b–ii, c–iv, d–iii

(3) a–iii, b–ii, c–i, d–iv

(4) a–ii, b–i, c–iii, d–iv

53)

	Column I		Column II
(a)	Earthworm	(i)	Nephridia
(b)	Cockroach	(ii)	Malpighian tubules
(c)	Prawn	(iii)	Flame cells
(d)	Planaria	(iv)	Green glands

(1) a–i, b–ii, c–iii, d–iv

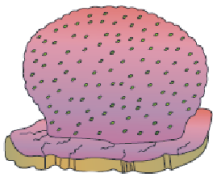
(2) a–ii, b–i, c–iii, d–iv

(3) a–iii, b–ii, c–i, d–iv

(4) a–i, b–ii, c–iv, d–iii

54) Which one is correct match in the following table regarding their name, phylum and characters of animals

	Figure	Name	Phylum	Characters

(1)		Euspongia	Porifera	(i) water canal system (ii) skeleton is made up of spongin fibres
(2)		Aurelia	Coelenterata	(i) medusa (ii) tissue level organisation and diploblastic
(3)		Cnidoblast	Ctenophora	(i) stinging cell (ii) used for anchorage defence and capture of prey
(4)		Pleurobrachia	Ctenophora	(i) 8 external row of ciliary comb plate (ii) Fertilisation external

(1) 1

(2) 2

(3) 3

(4) 4

55) **Statement I:** Metamerism is seen in earthworms.
Statement II: Notochord is present in all non-chordates.
Statement III: Arthropods have closed circulatory system.
Statement IV: Annelida have chitinous exoskeleton.

(1) statement I and II are false.

(2) statement II and IV are true.

(3) statement II and III are true.

(4) statement II,III and IV are false.

56) Match the Column:

Column-I		**Column-II**	
A.	Arthropoda	1.	Gills
B.	Mollusca	2.	Book lungs

C.	Echinodermata	3.	Water vascular system
D.	Hemichordata	4.	Gills

(1) A-3, B-4, C-2, D-1

(2) A-4, B-1, C-3, D-2

(3) A-2, B-1, C-3, D-4

(4) A-2, B-4, C-2, D-3

57) Match coelom type:

Column-I		**Column-II**	
A.	Platyhelminthes	1.	Pseudocoelomate
B.	Aschelminthes	2.	Enterocoelomate
C.	Echinodermata	3.	Acoelomate
D.	Arthropoda	4.	Schizocoelomate

(1) A-3, B-1, C-2, D-4

(2) A-2, B-1, C-3, D-4

(3) A-3, B-1, C-4, D-2

(4) A-3, B-4, C-2, D-1

58) Match the columns and choose the correct answer from the options given below

	Phylum	**Common name of animal**	**Scientific name of animal**
a	Platyhelmintehs	(i) roundworm	(u) *Taenia*
b	Aschelminthes	(ii) earthworm	(v) *Pheretima*
c	Annelida	(iii) tapeworm	(w) *Ascaris*

	a	**b**	**c**
1	ii,v	i,w	iii,u
2	iii,u	ii,w	i,v
3	iii,u	ii,v	i,w
4	iii,u	i,w	ii,v

(1) 1

(2) 2

(3) 3

(4) 4

59) Match the following characteristic with respective animal :

	Column-I		**Column-II**
(A)	Radula	(I)	Planaria

(B)	Muscular pharynx	(II)	Limulus
(C)	Abscence of Anus	(III)	Pila
(D)	Jointed appendages	(IV)	Wuchereria

(1) A-IV, B-III, C-I, D-II

(2) A-III, B-IV, C-I, D-II

(3) A-III, B-I, C-IV, D-II

(4) A-II, B-III, C-I, D-IV

60) **Statement-I :-** Inner wall of sponge body helps in water current formation.
Statement-II :- Flagellated choanocytes circulates water in spongocoel from ostia to osculum.

(1) Statement II is incorrect

(2) Statement I is correct only

(3) Statement I & II both are incorrect

(4) Statement I & II both are correct

61) (A) The body is covered by chitinous exoskeleton.
(B) They are mostly monoecious.
(C) They are bilaterally symmetrical, triploblastic and coelomate animals.
(D) They have open blood circulation usually.
Choose the correct statements about Arthropodes.

(1) A only

(2) A and B only

(3) A, B and D only

(4) A, C and D only

62) Mantle cavity is found between :-

(1) Mantle and shell

(2) Hump and mantle

(3) head and muscular foot

(4) Radula and muscular foot

63) While studying a soft-bodied animal, a student observed that the body is dorsoventrally flattened, triploblastic, and no body cavity is present. What is the coelomic condition of this animal?

(1) Pseudocoelomate

(2) Coelomate

(3) Acoelomate

(4) Schizocoelomate

64) **Assertion (A):** Nematodes are pseudocoelomates where body cavity is not completely lined by mesoderm.
Reason (R): In pseudocoelomates, mesoderm is present only towards the body wall but absent towards the alimentary canal.

Choose the correct answer:

(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true but R is not the correct explanation of A

(3) A is true but R is false

(4) A is false but R is true

65) **Statement I :** Sponges belonging to Porifera possess a water canal system where water enters through ostia, circulates through spongocoel and exits via osculum, facilitating respiration, excretion and food capture.
Statement II : Poriferans have a cellular level of organization without true tissues and their body is supported by spicules or spongin fibres.
Choose the correct option:

(1) Both statements are true and Statement II explains Statement I

(2) Both statements are true but Statement II does not explain Statement I

(3) Statement I is true but Statement II is false

(4) Statement I is false but Statement II is true

66) 'K' represents a group of animals having porous body, canal system and cellular level of organization. They lack true tissues and organs. Based on skeletal elements and canal system, 'K' is divided into different classes. Water current plays a significant role in their life processes.
Select the correct sequence of water flow in 'K':
W - Spongocoel
X - Ostia
Y - Osculum

(1) X → W → Y

(2) X → Y → W

(3) Y → W → X

(4) W → X → Y

67) Identify the statement that is NOT correct.

(1) Echinoderms are exclusively marine animals.

(2) Adults show radial symmetry while larvae are bilateral.

(3) Water vascular system helps in locomotion and feeding.

(4) They possess a closed circulatory system.

68) Which is not found in bacteria ?

(1) Mitochondria

(2) DNA

(3) Ribosome

(4) Cytoplasm

69) According to the size which sequence is not correct :

(1) Eukaryotic cell > Prokaryotic cell > PPLO > Virus
(2) Virus < PPLO < Prokaryotic cell < Eukaryotic cell
(3) Eukaryotic cell > PPLO > Prokaryotic cell > Virus
(4) PPLO < Prokaryotic cell < Eukaryotic cell < Tissue

70) Who saw and described 1st live cell ?

(1) Robert Hooke
(2) Flemming
(3) Anton Van Leeuwenhoek
(4) Schwann

71)

Who was the one to describe that cells divided and new cells are formed from pre-existing cells?

(1) Schleiden
(2) Schwann
(3) Virchow
(4) All of these

72)

'Omnis cellula e cellula' was given in the year _____

(1) 1756
(2) 1855
(3) 1945
(4) 1839

73)

Who formulated the cell theory?

(1) Schleiden
(2) Schwann
(3) Both (1) and (2)
(4) Robert Brown

74) A plant cell has :–

(1) a large central vacuole and rigid cell wall
(2) a centriole for cell division
(3) a centrosome inactive in non-dividing cells
(4) absence of cell membrane

75) Which of the following is main arena of cellular activities which maintains a cell in "living state"

?

(1) Mitochondria
(2) Cell membrane
(3) Nucleus
(4) Cytoplasm

76) Select the wrong pair :-

Types of cells		**Shape of cells**
(1)	Red blood cells	Round and biconcave
(2)	Tracheid cells	Elongated
(3)	Nerve cell	Unbranched and long
(4)	Mesophyll cells	Round and oval

(1) 1
(2) 2
(3) 3
(4) 4

77) "Animalcule" term was given by :-

(1) Leeuwenhoek
(2) Robert Hooke
(3) Robert Brown
(4) Rudolf Virchow

78)

Match the following columns :-

	Column-I		**Column-II**
(a)	Anton Von Leeuwenhoek	(i)	Cork cell
(b)	Robert Brown	(ii)	Live cell
(c)	Robert Hook	(iii)	*Omins cellula-e-cellula*
(d)	Rudolf Virchow	(iv)	Nucleus

(1) (a)(ii), b(iv), c(i), (d)(iii)
(2) (a)(i), b(ii), c(iii), (d)(iv)

(3) (a)(ii), b(i), c(iii), (d)(iv)

(4) (a)(iii), b(ii), c(i), (d)(iv)

79)

The cell wall of a young plant cell, the ....."A"..... is capable of growth, which gradually diminishes as the cell mature and the ....."B"..... is formed on the inner side of the cell. Choose the correct word for "A" and "B" respectively :-

(1) Secondary wall, Tertiary wall

(2) Primary wall, Secondary wall

(3) Secondary wall, Primary wall

(4) Tertiary wall, Secondary wall

80) Select the incorrect statement :

(1) Anything less than a complete structure of cell does not ensure independent living

(2) Anton Von Leeuwenhoek first saw and described a living cell

(3) Robert Brown discovered cell

(4) Cell is the basic unit of structure and function of all organisms

81) Given below are two statements : One is labelled as Assertion (A) and the other is labelled as Reason (R)
Assertion (A) : Peripheral proteins are partially or totally buried in the plasma membrane.
Reason (R) : The quasi fluid nature of lipid enables lateral movement of proteins with in the over all bilayer,
In the light of the above statements, choose the correct answer from the option given below :

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

(2) (A) is correct but (R) is not correct

(3) (A) is not correct but (R) is correct

(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

82) Match the following :-

(a)	Simple diffusion	(I)	Water
(b)	Osmosis	(II)	Polar molecules
(c)	Active transport	(III)	Neutral -solutes
(d)	Facilitated diffusion	(IV)	Na^+ – K^+ pump

(1) a–I, b–III, c–II, d–IV

(2) a–I, b–IV, c–III, d–II

(3) a–III, b–I, c–IV, d–II

(4) a–III, b–I, c–II, d–IV

83)

Which one of the following statements refers to reductionist biology ?

(1) Behavioural approach to study and understand living organisms

(2) Physico-chemical approach to study and understand living organisms.

(3) Physiological approach to study and understand living organisms.

(4) Chemical approach to study and understand living organisms.

84) Select the incorrect statement :

(1) Phospholipid is the main component of cell membrane

(2) Phospholipid layer provides fluidity to plasma membrane because phospholipids are rich in saturated fatty acids

(3) The Quasifluid nature of lipid enable lateral movement of protein with in the overall bilayer

(4) The fluid nature of the membrane is also important in various functions like cell growth, formation of intercellular junction, endocytosis, secretion, cell division etc.

85) How many of the following is/are correct statement regarding cell membrane ?
(A) The carbohydrate component of cell membrane mainly consist of cellulose
(B) Watson and crick proposed fluid mosaic model.
(C) Lipids are arranged in a bilayer with polar head towards the outer side
(D) Na^+ and K^+ ion move across cell membrane by only passive transport.
Option :-

(1) One

(2) Two

(3) Three

(4) Four

86) Which of the following statement is not correct-

(1) Hopanoid provide the stability to the prokaryotic cell membrane

(2) Plasma membrane have enzymes

(3) Plasma membrane in an asymmetric structure

(4) Tunnel protein may be integral or peripheral protein

87) Read the following four statement (a-d)
(a) The quasifluid nature of lipid enables lateral movement of protein with in the overall bilayer.
(b) Membrane of R.B.C. has approximately 52% protein and 40% lipid
(c) Neutral solutes may move across the membrane by the process of simple diffusion.
(d) Flip flop movement are shown by both lipid and protein.
How many of the above statements are right?

(1) Three

(2) Four

(3) Two

(4) One

88) Cell wall is :-

(1) Non living and semipermeable

(2) Non-living and permeable

(3) Living and selectively permeable

(4) Living and impermeable

89) **Statement-I :** Cell membrane show fluidity.
Statement-II : Fluidity is due to presence of proteins.

(1) Both statements are correct.

(2) Statement-I is correct and Statement-II incorrect.

(3) Statement-I is incorrect and statement-II is correct.

(4) Both statements are incorrect.

90) In eukaryotic cells, why there is a extensive compartmentalization of cytoplasm?

(1) Due to the presence of ribosome

(2) Due to the absence of organelles

(3) Due to the presence of membranous organelles

(4) All of the above

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	1	2	2	2	1	3	2	1	4	3	2	2	1	4	1	3	1	4	3	3
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	1	1	2	2	2	2	3	4	1	1	3	3	1	2	3	2	3	4	4	1
Q.	41	42	43	44	45															
A.	4	1	3	3	1															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	2	4	3	1	2	1	4	2	2	2	1	3	2	4	3	4	4	4	2	4
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	1	2	4	2	1	2	2	1	2	2	1	4	4	4	1	4	3	1	1	4
Q.	86	87	88	89	90															
A.	3	3	3	1	2															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	3	1	4	3	4	3	2	3	4	1	4	4	3	1	3	1	2	1	2	1
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	4	3	3	3	2	3	2	4	3	3	3	4	1	2	3	4	3	3	3	2
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	2	1	3	2	4	4	3	4	1	2	2	4	4	4	3	1	4	2	4
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	4	2	3	1	2	1	4	1	3	3	3	2	3	1	4	3	1	1	2	3
Q.	171	172	173	174	175	176	177	178	179	180										
A.	3	3	2	2	1	4	1	2	2	3										

SOLUTIONS

PHYSICS

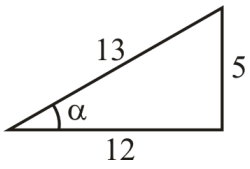
1)

$1 - \sin^2 37 - \cos^2 37 + 1$
$= 2 - (\sin^2 37 + \cos^2 37)$
$= 2 - 1 = 1$

2) $\cos 74° = \cos (2 \times 37°)$
$= \cos 2\ 37° - \sin 2\ 37°$
$= \left(\frac{4}{5}\right)^2 - \left(\frac{3}{5}\right)^2$
$= \frac{16}{25} - \frac{9}{25} = \frac{7}{25}$

3) Angle $(\theta) = \frac{\text{arc}}{\text{radius}} = \frac{15\text{ cm}}{5\text{ cm}} = 3$ rad

4) $\frac{x}{2} + \frac{y}{4} = 1$
$\Rightarrow y = -2x + 4$
$\Rightarrow$ y intercept = 4

5) 

$\sin\alpha = \frac{5}{13}$
$\sin\alpha = 2\sin\alpha \cos\alpha$
$= 2\left(\frac{5}{13}\right)\left(\frac{12}{13}\right) = \frac{120}{169}$

6) $\sqrt{0.999} = (1 - 0.001)^{1/2}$
By Binomial theorem
$= 1 - \frac{1}{2}(0.001)$
$= (1 - 0.005)$
$= 0.9995$

7)

Least value $= -\sqrt{4^2 + 3^2}$

$= -5$

8)

$y = \frac{4}{3}x + 4 \;=\; m\,x + c$

9) $\frac{1}{2}\int_0^{\pi/2} (1 - \cos 2\theta)\, d\theta$

$= \frac{1}{2}\left[\int_0^{\pi/2} d\theta - \int_0^{\pi/2} \cos 2\theta \; d\theta\right] = \frac{\pi}{4}$

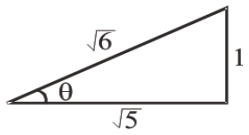
10) $\cos\left(12\pi + \frac{\pi}{6}\right) = \cos\frac{\pi}{6} = \frac{\sqrt{3}}{2}$

11) $S = \frac{1}{1 - \left(-\frac{1}{2}\right)} = \frac{2}{3}$

12) $\alpha = 6,\ \beta = 4$

$(\alpha - \beta) = 2$

13)



$\Rightarrow \cos\theta = \frac{\sqrt{5}}{\sqrt{6}}$

14) line (1) has positive y intercept

15) $\int_{n-1}^{n} (3 + 4t)\, dt = \left(3t + 2t^2\right)_{n-1}^{n}$

$= \left(3t + 2t^2\right)_{n-1}^{n} = \left(3n + 2n^2\right) - \left[3(n-1) + 2(n-1)^2\right] = 4n + 1$

16)

By observation

17) $x = t^2 \Rightarrow \frac{dx}{dt} = 2t$ which is straight line passing through origin.

18) $y = \sin(x^2 - 4)$

$\frac{dy}{dx} = \cos(x^2 - 4)\frac{d}{dx}(x^2 - 4)$

$\frac{dy}{dx} = 2x \cdot \cos(x^2 - 4)$

19) $y = x^2 - 10x + 40$

$\frac{dy}{dx} = 2x - 10$

for $y_{min} \Rightarrow \frac{dy}{dx} = 0$

$\Rightarrow x = 5$

at x = 5, y will be y = 15

20) $v = a^3 \frac{dv}{dt} = 3a^2 \frac{da}{dt}$

$\frac{dv}{dt} = 3 \times 22 \times (0.1 \times 10^{-3})$

$\frac{dv}{dt} = 12 \times 10^{-4}\ m^3\ sec^{-1}$

21) $\int_R^\infty \frac{GMm}{x^2} dx = GMm \int_R^\infty \frac{dx}{x^2}$

$= GMm \left(-\frac{1}{x}\right)_R^\infty$

$= -GMm \left(\frac{1}{\infty} - \frac{1}{R}\right) = \frac{GMm}{R}$

22) $\int_0^{\pi/2} \sin x dx = [-\cos x]_0^{\pi/2}$

$= -(0 - 1) = 1$ unit

23)

$R = K \sin 2\theta$

$\frac{dR}{d\theta} = 2K \cos 2\theta = 0$

$\cos 2\theta = 0 \Rightarrow 2\theta = \frac{\pi}{2}$

$\theta = \frac{\pi}{4}$

24)

$12y = x^3$

$$12\frac{dy}{dt} = 3x^2\frac{dx}{dt}$$

$$\frac{dy}{dt} = \left(\frac{x^2}{4}\right)\frac{dx}{dt}$$

at x = 10

$$\frac{dy}{dt} > \frac{dx}{dt}$$

⇒ y-coordinate change at faster rate than π-coordinate

25)

$y^2 = 2x$

$\Rightarrow y^2 \propto x$

$y \propto \sqrt{x}$

represent straight line

26)

Explanation of question: The graph shows a semicircular curve on the x-axis. The question asks how the magnitude of slope changes as we move along the curve from left to right.

Concept: Magnitude of slope indicates how steep or flat the curve is at different points on it.

Solution: At the left end, the curve is very steep so the magnitude of slope is large. Moving toward the top, the curve becomes flatter so the magnitude decreases. After crossing the top, the curve again becomes steeper toward the right end, so the magnitude increases

Final Answer: Option (2)

27)

V = (2t + 3) m/s

$$V_{avg} = \frac{\int_0^1 V dt}{\int dt} = \frac{\int_0^1 (2t+3)dt}{1}$$

$V_{avg} = (t^2 + 3t)_0^1 = 4$ m/s

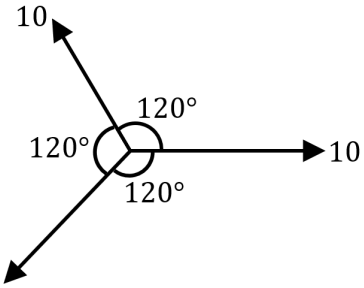
28) $\cos\theta = -\frac{P}{Q}$

$$R = \sqrt{P^2 + Q^2 + 2PQ\cos\theta}$$

$$\frac{Q^2}{4} = P^2 + Q^2 + 2PQ\left(-\frac{P}{Q}\right)$$

$$\frac{Q^2}{4} = Q^2 - P^2 \Rightarrow P^2 = \frac{3}{4}Q^2 \Rightarrow \frac{P}{Q} = \frac{\sqrt{3}}{2}$$

$\cos\theta = -\frac{\sqrt{3}}{2} \Rightarrow \pi = 150°$



29)

30) $\vec{R} = \left(\vec{A} + \hat{B}\right) + \left(\vec{A} - \hat{B}\right) \Rightarrow \vec{R} = 2\vec{A}$

Thus $\vec{R}$ and $\vec{A}$ are in same direction.

31) $\vec{R} = -\hat{i} + 5\hat{j} - 3\hat{k}$

$\hat{n} = \frac{\vec{R}}{|\vec{R}|} = \frac{-\hat{i} + 5\hat{j} - 3\hat{k}}{\sqrt{35}}$

32) $|\vec{a} + \vec{b}| = |\vec{a} - \vec{b}|$

$\sqrt{a^2 + b^2 + 2ab\cos\theta} = \sqrt{a^2 + b^2 - 2ab\cos\theta}$

Square both the side

$a^2 + b^2 + 2ab\cos\theta = a^2 + b^2 - 2ab\cos\theta$

$4ab\cos\theta = 0 \Rightarrow \cos\theta = 0 \Rightarrow \theta = 90°$

33) When drawn as per polygon law of vector addition, then will form a closed regular polygon. Hence resultant will be zero.

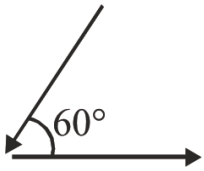
34) **Question Explaining :**

Find resultant of two vectors.

Concept :

This question is based on Addition of two vectors.

Solution :



$R = |\vec{A} + \vec{B}| = \sqrt{A^2 + B^2 + 2AB\cos\theta}$

$= \sqrt{7^2 + 7^2 + 2 \times 7 \times 7\cos 120}$

$= 7$

Final Answer : (2)

35) $R = \sqrt{9P^2 + 4P^2 + 12P^2\cos\theta}$

$2R = \sqrt{36P^2 + 4P^2 + 24P^2\cos\theta}$

$\cos\theta = -\frac{1}{2}$

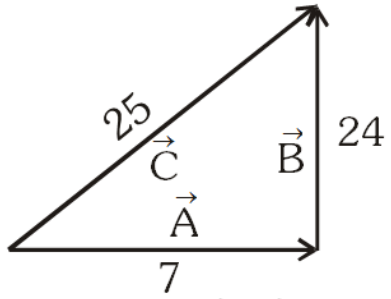
$\theta = 120°$

36)

Distance = $|\vec{r}_2 - \vec{r}_1| = \left|-4\hat{i} - 3\hat{j}\right| = \sqrt{(-4)^2 + (-3)^2}$

$= \sqrt{25} = \mathbf{5\ m}$

37) $\vec{A} + \vec{B} = \vec{C}$ & $A^2 + B^2 = C^2$



Angle between $\vec{A}$ & $\vec{B}$ is 90°

38) If $\left|\vec{A} + \vec{B}\right| = \left|\vec{A}\right| = \left|\vec{B}\right|$

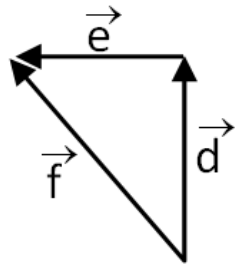
then angle between them $\theta = 120°$

Let $\left|\vec{A}\right| = \left|\vec{B}\right|$

So $\left|\vec{A} - \vec{B}\right| = \sqrt{A^2 + B^2 - 2AB\cos\theta}$

$= \sqrt{a^2 + a^2 - 2a^2\left(-1/2\right)} = \sqrt{3}\left|\vec{A}\right|$

39)



$\Rightarrow \vec{f} = \vec{d} + \vec{e}$

40) $\left(\sqrt{(x^2 + y^2)}\right)^2$

$$= \left(\sqrt{(x+y)^2 + (x-y)^2 + 2(x+y)(x-y)\cos\theta}\right)^2$$

$x^2 + y^2 = 2(x^2 + y^2) + 2(x^2 - y^2)\cos\theta$

$$\cos\theta = -\frac{\left(x^2+y^2\right)}{2\left(x^2-y^2\right)}$$

41) $|A-B| \leqslant R \leqslant (A+B)$

42) **Explanation Question:** Differentiation of different functions
Concept: This question is based on differentiation
Solution: (P) $\frac{d(x\ell nx)}{dx} = \frac{xd\ell nx}{dx} + \ell nx\frac{dx}{dx}$

$= x\frac{1}{x} + \ell nx$

$= 1 +$ n x

(S) $y = \frac{x}{2x+1} \Rightarrow \ell ny = nx - \ell n(2x+1)$

$\frac{1}{y}\frac{dy}{dx} = \frac{1}{x} - \frac{4}{2x+1} = \frac{1}{x(2x+1)}$

$\frac{dy}{dx} = \frac{1}{(2x+1)^2}$

(R) $y = 6x^2 \Rightarrow \frac{dy}{dx} = 12x$

$\Rightarrow \frac{d^2y}{dx^2} = 12$

Final Answer
The correct option is (1)

43) **Explain Question:**
Concept:
Solution:

$v = \sqrt{\sin 2t}$

$\frac{dv}{dt} = \frac{1}{2\sqrt{\sin 2t}}.\cos 2t.2$

$a = \frac{dv}{dt} = \frac{\cos 2t}{\sqrt{\sin 2t}}$

$at\frac{\pi}{y}$

$a = \frac{\cos\frac{\pi}{2}}{\sqrt{\sin\frac{\pi}{2}}} = 0$

Final answer: Acceleration at time t $= \frac{\pi}{4}$ is 0 option 3

44) **Question Explanation:** The question asks to find the derivative.
Concept: Differentiation of Polynomial and Trig Functions.
Solution :

$\frac{d}{dx}\left(3x^3 + 4\sin x - 3\cos x\right)$

$\frac{d}{dx}\left(3x^3\right) + \frac{d}{dx}(4\sin x) + \frac{d}{dx}(-3\cos x)$

$= 4x^2 + 4\cos x + 3\sin x$

Final Answer : The correct option is (3)

45) $\frac{dx}{dt} = 6t \qquad \frac{dy}{dt} = 6t^2$

$\therefore \quad \frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{6t^2}{6t} = t$

CHEMISTRY

46) Gold foil experiment

47) Charge/mass for n = 0, for $\alpha = \frac{2}{4}$,
for $p = \frac{1}{1}$, for $e^- = \frac{1}{1/1837}$

48)

$E = \frac{hc}{\lambda} \Rightarrow E \propto \frac{1}{\lambda}$

$\frac{E_1}{E_2} = \frac{\lambda_2}{\lambda_1} = \frac{6000}{3000}$

$= 2 : 1$

49)

$^{238}_{92}U$ is isodiapher with $^{234}_{Z}X$.

$^{238}_{92}U$

neutron (n) = A – P = 238 – 92 = 146

n – p = 146 – 92 = 54

Isodiapher means (n – p) remain same.

For $^{234}_{Z}X$

n – p = 54

n + p = 234

\+ _____

2n = 288

n = 144

n – p = 54

144 – p = 54

p = 144 – 54 = 90

50) Each K^+, Cl^-, Ca^{2+}, S^{2-} has 18 electrons.

51) % change in mass

$$= \frac{\text{final mass} - \text{initial mass}}{\text{initial mass}} \times 100$$

52)

$$\text{Average weight} = \frac{W_1X_1 + W_2X_2}{X_1 + X_2}$$

Here $X_1 + X_2 = 100$

53) Energy of one photon = $\frac{hc}{\lambda}$

Lets assume 'n' no. of photons are present :

$$\therefore n \times \frac{hc}{\lambda} = 3 \text{ J}$$

$$\Rightarrow n \times \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{662.6 \times 10^{-9}} = 3$$

$$\therefore n = 10^{19}$$

54)

$$E_{absorbed} = E_{emitted}$$

$$\frac{hc}{\lambda} = \frac{hc}{\lambda_1} + \frac{hc}{\lambda_2}$$

$$\frac{1}{\lambda} = \frac{1}{\lambda_1} + \frac{1}{\lambda_2}$$

$$\lambda = \frac{\lambda_1\lambda_2}{\lambda_1 + \lambda_2}$$

55) No of electrons = z – (change with sign)
= 7 – (–3) = 10

56)

Electron–nucleus attraction → potential energy becomes **more negative** (i.e., decreases).

As attraction increases → electron speeds up → kinetic energy **increases**.

Total energy = kinetic energy + potential energy → becomes **more negative** overall → **decreases**.

Close to nucleus:

Kinetic energy ↑

Potential energy ↓ (more negative)

Total energy ↓

57) **Assertion (A):** Few positively charged α-particles were deflected → True, due to repulsion from the dense, positively charged nucleus.

Reason (R): Positive charge is spread throughout the atom → False, this was the Thomson model, which Rutherford disproved.

Hence, A is true but R is false.

58) Let fraction of M^{50} = x, so fraction of M^{52} = 1 – x.
Average atomic weight = 50x + 52(1 – x)
51.7 = 50x + 52 – 52x
51.7 = 52 – 2x
2x = 52 – 51.7 = 0.3
x = 0.15
So percentage occurrence of M^{50} = 15%

59)

• Radiowaves have the longest wavelength → lowest frequency.
• Infrared (R) shorter wavelength than radiowaves → higher frequency
• Ultraviolet (UV) shorter wavelength than IR → higher frequency
• X-rays have the shortest wavelength → highest frequency
Therefore X-rays have the maximum frequency.

60)

$$\bar{\nu} = \frac{1}{\lambda} = \frac{1}{5 \times 10^{-9} m} = 2 \times 10^{8} m^{-1}$$

61) Option 1: Correct. The wavelength is defined as the distance between two consecutive peaks (crests) or troughs.
Option 2: Correct. Frequency is the number of wave cycles that pass a fixed point in one second.
Option 3: Correct. Velocity (v) represents the distance a wave travels per unit of time (typically 3 x 10^8 m/s in a vacuum).
Option 4: Incorrect. While all electromagnetic waves travel at the same speed (the speed of light), they have different wavelengths and frequencies, which characterize their position in the spectrum (e.g., Radio waves vs. X-rays).

62)

The relation between frequency (v), wavelength (λ) and speed of light (c) is :
$v = c/\lambda$
Where :
v = frequency of radiation (Hz)
c = speed of light = 3×10^8 m/s
λ = wavelength of radiation
Given :
$l = 1Å = 1 \times 10^{-10}m$
Substitute the values :
$v = (3 \times 10^8)/(1 \times 10^{-10}) = 3 \times 10^{18}$ Hz
Since 1 Hz = $1s^{-1}$ = 1 cycle/sec, all representations are equivalent.

63)

$$r \propto \frac{n^2}{Z}$$
$r_1 : r_2 : r_3 = 1 : 4 : 9$

64)

For Lyman series, λ_{min}, $n_1 = 1$ and $n_2 = \infty$ and for λ_{max} $n_1 = 1$ and $n_2 = 2$
$$\frac{1}{\lambda} \propto \left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$
$$\left(\frac{1}{\lambda}\right)_{min} \propto \left(\frac{1}{1^2} - \frac{1}{\infty}\right) \propto 1$$
$\lambda_{min} \propto 1$
$$\left(\frac{1}{\lambda}\right)_{max} \propto \left(\frac{1}{1^2} - \frac{1}{2^2}\right) \propto \frac{3}{4}$$
$\lambda_{max} \propto \frac{4}{3}$
$$\frac{\lambda_{min.}}{\lambda_{max.}} = \frac{1}{\left(\frac{4}{3}\right)} = \frac{3}{4}$$ $\Rightarrow 3:4$
Final Answer : Option (2)

65)

$\square = 3 \rightarrow m = -3$ to $+3 \rightarrow$ total values $= 2\square + 1 = 7$
Max $e^- = 2 \times 7 = 14$.

66)

Mn (Z = 25) = $1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^5$
$2s^2 \rightarrow$ contains 2 electrons.

67)

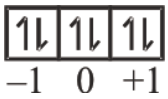
Possible (n, $\square$) with $n + \square = 6$:

• (6, 0) → 6s orbital, m = 0 → 1 orbital
• (5, 1) → 5p orbital, m = 0 → 1 orbital
• (4, 2) → 4d orbital, m = 0 → 1 orbital
Total = 3 orbitals.

68)

(a)

Mg =	$1s^2$	$2s^2$	$2p^6$	$3s^2$	
ℓ =	0	0	1	0	
m =	0	0	1	0	for m = 0
					No. of e^- = 8



(b) One orbital occupied maximum $2e^-$
(c) M shell n = 3, no. of $e^- = 2n^2 = 2 \times 3^2 = 18$
(d) $P = 1s^2 2s^2 2p^6 3s^2 3p^3$

l = 0 for s orbital, so $6e^-$, in s, so for 3e, s = $+\frac{1}{2}$

and for 3e, s = $-\frac{1}{2}$

69)

In Bohr model, velocity of electron is inversely proportional to principal quantum number ($v \propto 1/n$).
For first orbit: $v_1 = x$ when $n_1 = 1$
For third orbit: $n_2 = 3$
$v_2 / v_1 = n_1 / n_2 = 1 / 3$
$v_2 = x \times (1/3) = x/3$

70) Standard formula: $R = R_0 \times A^{1/3}$, where $R_0 \approx 1.33 \times 10^{-15}$ m
Implies : nuclear radius cube root of mass number

71) 16 amu → 100
12 amu → $\frac{100}{16} \times 12$
= 75

72)

1. n = 3, □ = 2, m = +1, S = –1/2 → Valid (m within –2 to +2)
2. n = 4, □ = 1, m = +2, S = +1/2 → Not valid (m must be –1, 0, +1)
3. n = 4, □ = 3, m = 0, S = +1/2 → Valid (m within –3 to +3)
4. n = 3, □ = 1, m = 0, S = –1/2 → Valid

73)

Sixth excited state ⇒ n = 1 + 6 = 7
Number of de-Broglie waves = n = 7

74)

PE = 2TE

75)

E_4 = –0.85 eV in H-atom.

76)

Calculate Z/n for each case:
1. Li^{2+}, n = 3 → Z/n = 3/3 = 1
2. Be^{3+}, n = 6 → Z/n = 4/6 ≈ 0.667
3. He^{+}, n = 8 → Z/n = 2/8 = 0.25
4. H, n = 4 → Z/n = 1/4 = 0.25
highest Z/n → Li^{2+}, n = 3 → V_{max}

77) $mvr = \frac{nh}{2\pi}$

78) the order of energy of orbitals can be calculated from (n + ℓ) rule. The lower the value of (n + ℓ) for an orbital, lower is its energy. If two orbitals have same (n + ℓ) value, the orbital with lower value of n has the lower energy.

79)

For 3p orbital : n = 3, ℓ = 1
Radial nodes = 3 – 1 – 1 = 1
Final Answer : 1

80)

For single electron species 1s < 2s = 2p < 3s = 3p = 3d < .....

81) $\lambda = \frac{h}{mv}$
$= \frac{6.6 \times 10^{-34}}{0.66 \times 100}$

82) • Number of subshells:
Number of subshells $= n$
For n = 3 :
Number of subshells $= 3$ (3s, 3p, 3d)
• Number of orbitals:
Number of orbitals $= n^2$
For n = 3 :

Number of orbitals $= 3^2 = 9$

83)

Mass of bullet = 10g = 0.01 kg
$\Delta x = 10^{-5}$ m
Planck's constant h = 6.626×10^{-34} J-s

$$\Delta V = \frac{6.626 \times 10^{-34}}{4\pi \cdot 0.01 \cdot 10^{-5}}$$

$$\Delta V = \frac{6.626 \times 10^{-34}}{1.256 \times 10^{-7}} \approx 5.28 \times 10^{-27} \text{m/s}$$

84)

For Be^{+3}, the atomic number (Z) is 4.
The third excited state corresponds to the principal quantum number n = 4.

Using the energy formula:

$$E_4 = -13.6 \times \frac{Z^2}{n^2}$$

$$E_4 = -13.6 \times \frac{4^2}{4^2}$$

$$E_4 = -13.6 \times \frac{16}{16}$$

$E_4 = -13.6$ eV

85)

Copper (Cu) atomic number = 29
Electronic configuration: [Ar] $3d^{10}$ $4s^1$
Cu^{2+} loses 2 electrons → remove from 4s first, then 3d
Cu^{2+} configuration: [Ar] $3d^9$
A 3d subshell can hold 10 electrons; $3d^9$ has 1 unpaired electron

86) Violation of Hund's rule means electrons are paired in an orbital before all orbitals in the subshell have at least one electron with the same spin.
An electron distribution where all orbitals in a subshell have at least one electron, but one orbital has two electrons while others have only one with the same spin, violates only Hund's rule.

87)

Explain the Question : The question requires us to identify which of the given orbital diagrams simultaneously violates two fundamental principles of electron configuration: Pauli's exclusion principle and Hund's rule.
Concept : This question is based on pauli's exclusion principle states that an orbital can hold a maximum of two electrons, and they must have opposite spins, while Hund's rule dictates that

degenerate orbitals must be filled with one electron each before any are completely filled.
Solution :
Let's examine the orbital diagram in option 3 : [↑ ↓] [↑ ↑] [↑] [ ]
Violation of pauli's exclusion principle : The first p-orbital, represented as [↑ ↑], contains two electrons with the same spin (both pointing up). This is a direct violation of Pauli's principle.
Violation of Hund's Rule : The electrons are being paired up in the first p-orbital before all the degenerate p-orbitals are occupied by at least one electron. Specifically, the third p-orbital is empty while the first is being filled with a second electron, which violates Hund's rule.
Final Answer : 3

88)

Explain Question : We need to match each concept in Column-I with its correct description/formula from Column-II

Concept : Basic quantum mechanical principles and laws.

Solution :
(i) Aufbau principla → (d) Electronic configuration (electrons fill orbitals in increasing energy)
(ii) De Broglie's hypothesis → (a) $\lambda = h/(mv)$ (matter waves)
(iii) Hund's rule → (c) Max. unpaired e^- (maximum multiplicity)
(iv) Planck's law → (b) $\lambda = hc/\Delta E$ (energy-frequency relation)

(i) → (d), (ii) → (a), (iii) → (c), (iv) → (b)

Final Answer : option (3)

89) $TE = -KE = \frac{1}{2}PE$

$PE_1 = 2TE_1 = 2 \times -13.6\ eV = -27.2\ eV$ (First orbit)

$$PE_3 = 2TE_3 = 2 \times \frac{-13.6}{9} = \frac{-27.2}{9} eV$$

$$PE_3 - PE_1 = \frac{-27.2}{9} - (-27.2)$$

$$= 27.2 \times \frac{8}{9} = 24.18 eV$$

90) $E \propto \frac{1}{n^2}$ $\quad \frac{E_3}{E_2} = \frac{(2)^2}{(3)^2} = \frac{4}{9}$

BIOLOGY

91)

Question Explanation: Identify the correct statement regarding compound epithelium.

Concept: Compound epithelium

Solution: The correct answer is option 3

A. Compound epithelium consists of multiple layers of cells and mainly performs a protective function, not secretion or absorption.
B. It is found in areas subjected to wear and tear such as the epidermis of skin (dry surface) and lining of buccal cavity (moist surface).

Final Answer: Option 3

92)

Question Explanation: We need to identify which statements correctly describe the epithelium shown, which is a stratified/compound epithelium (multi-layered).

Concept: Compound epithelium

Solution: A stratified (compound) epithelium is composed of more than one layer of cells, provides protection against chemical and mechanical stresses, and has limited role in secretion and absorption because of its multiple layers. It is not a simple epithelium (which has a single cell layer). So statements (a), (c), and (d) are correct about this epithelium.

Final answer: Option 1

93)

Bone matrix is non-pliable.

94)

Only some bone's Bone marrow produce blood cells.

95)

Module Pg. # 73, 74, 75

96) **Solution :** Reason (R): False → The frog's heart has 2 atria + 1 ventricle, so there is partial mixing of oxygenated and deoxygenated blood, not complete separation.
Final Answer : (3)

97)

Explain Question : Recognise the figure and find out the correct matching for 'a' and 'b'

Concept : This question is based on Phylum coelenterate.

Solution : Polyp produce medusa Asexually, and Medusa produce polyp sexually

Final Answer : option (2)

98)

NCERT Pg. # 40, 41, 42

99) NCERT XI Pg. # 49

100) **Explain Question:** Identify the incorrect statement about phylum Cnidaria.
Concept: This question based on Cnidaria
Solution: The correct answer is option 1.
Cnidarians possess cnidocytes containing stinging capsules called nematocysts, not "nematoblast" (which are developing cnidoblast cells).
Final answer: 1

101)

Question Explanation: Identify the option that is completely correct without any exception.

Concept: Animal Classification

Solution: The correct answer is option 4

A. Option 1
   Scypha (Porifera) does not show indirect development in the same way as Ctenophora, and fertilization pattern differs; not universally correct.

B. Option 2
   Taenia lacks an alimentary canal, though flame cells are present for osmoregulation. So the statement is incorrect.

C. Option 3
   Nereis has parapodia for locomotion, but Pheretima (earthworm) does not have parapodia; it moves using setae.

D. Option 4
   Pinctada and Pila are molluscs; both have a mantle layer covering the body and possess gills for respiration. This statement is correct without exception.

Final Answer: Option 4

102)

NCERT Pg. 39

103) NCERT (XI) (E) Pg. # 102

104) Conceptual.

105) Conceptual.

106) Conceptual.

107)

NCERT Pg. No. 103, 104

108)

NCERT Pg. No. 103, 104

109)

NCERT Pg. No. 103, 104

110)

NCERT Pg. No. 103, 104

111)

NCERT Pg. No. 103, 104

112)

NCERT Pg. No. 103, 104

113)

NCERT Pg. No. 103, 104

114)

NCERT Pg. No. 103, 104

115)

NCERT Pg. No. 103, 104

116)

Allen module

117)

Question Explanation
The question is asking to correctly match each organism with its characteristic feature based on their phylum and unique structures or properties.

Concept
Animal kingdom characteristic features

Solution
Fasciola is a platyhelminth and possesses flame cells for excretion.
Nereis is an annelid and shows parapodia for locomotion.
Hydra belongs to Cnidaria and has cnidoblast cells.
Comb jelly shows bioluminescence.

Final answer: Option 2

118) **Question Explanation:** Unique characteristic of Coelenterata.
Concept: Nematocysts
Solution: The phylum **Coelenterata** (Cnidaria) is uniquely characterized by the presence of stinging cells called **cnidoblasts** or **cnidocytes**. These cells contain a special capsule called a **nematocyst** , which is used for defense and for capturing prey. Other features like hermaphroditism and polymorphism are found in various other phyla as well.
Final Answer: Nematocysts

119) NCERT Pg. No. # 53

120) NCERT Page No. # 46, 47, 48

121) **Explain Question:** Correct feature of phylum Nematoda (Ascaris).
Concept: This question is based on Aschelminthes.
Solution:
The figure shows Ascaris, a roundworm belonging to phylum Aschelminthes. They have a complete alimentary canal with a muscular pharynx.
Final Answer: 3 – Alimentary canal is complete with a well-developed muscular pharynx.

122) **Explain Question:** Incorrect phylum–character matches.
Concept : This question is based on Non-Chordata.
Solution :
Hemichordata are marine, worm-like animals (e.g., Balanoglossus, Saccoglossus).They have a stomochord, but this is not a true notochord. So only match I is incorrect.
Final answer : Option 4 — Only I

123)

Question explanation:
The question is asking which set contains animals all belonging to the same phylum.
Concept: This question is based on Animal Kingdom.
Solution:
Option 1: Sycon, Euspongia, Spongilla – All are sponges, Phylum Porifera.
Final answer: (1)

124)

Question Explanation: The question asks the term used for serial repetition of body segments, each containing similar organs.

Concept: SEGMENTATION

Solution: Metamerism is the phenomenon in which the body of an organism is externally and internally divided into a series of similar, repeating segments, with each segment containing atleast some similar organs arranged serially.

Final answer: Option 2

125) NCERT (E) Pg. # 116

126) NCERT (XIth) Pg. # 119

127) NCERT Pg. # 118

128) NCERT Pg. # 120

129) NCERT (E) Pg. # 116

130) NCERT Pg. # 116, 7.5.1

131)

Question Explanation
The question is asking about the characteristics of a frog.

Concept
Animal physiology and classification

Solution
Frog excretes urea → ureotelic.
Frog shows differences between male and female → sexual dimorphism.
Frog eats insects → insectivorous.
Frog's body temperature depends on environment → cold-blooded (ectothermic).

Final answer

Option 4

132) NCERT (XIth) Pg. # 116

133) NCERT XI Page No. # 81

134) **Explanation:**
• **Hibernation:** This is a state of inactivity and metabolic depression that some animals enter during winter when food is scarce and temperatures are low. It's a survival strategy to conserve energy.
• **Camouflage:** This is the ability of an animal to blend in with its surroundings, making it difficult for predators or prey to spot them. This can be achieved through coloration, patterns, or behavior.
• **Poikilotherm:** These are animals whose body temperature varies with their surrounding environment. They are often referred to as "cold-blooded."
• **Aestivation:** This is a state of inactivity and metabolic depression that some animals enter during hot and dry periods. It's a survival strategy to conserve energy and water.
ans-3

135) NCERT Pg.# 116,121

136) Module Pg.# 159

137) NCERT Pg # 118, 119

138) NCERT Pg. # 50

139)

NCERT-XIth, Pg # 49

140) NCERT-XIth Pg. # 52

141) **Solution :** -
Final Answer : (2)

142) a–i, b–ii, c–iv, d–iii

143) a–i, b–ii, c–iv, d–iii

144) Conceptual

145) **Solution :** Metamerism is in annelids; notochord is in chordates, Notochord is present in chordates.
Arthropods have open circulatory system. Arthropods have chitinous exoskeleton.
Final Answer : (4)

146) **Solution :** Arthropoda book lungs, Mollusca gills, Echinodermata water vascular, Hemichordata gills.
Final Answer : (3)

147) **Solution :** Flatworms acoelomate, roundworms pseudocoelomate, annelids and arthropods coelomate.
Final Answer : (1)

148)

Question Explanation The question is asking to correctly match the phylum, common name, and scientific name of animals.

Concept Platyhelminthes, Aschelminthes and Annelida

Solution: Platyhelminthes → Tapeworm → Taenia
Aschelminthes → Roundworm → Ascaris
Annelida → Earthworm → Pheretima

So, a → iii,u ; b → i,w ; c → ii,v

Final answer: 4

149) NCERT, Pg # 52,43,54

150) NCERT, Pg # 49

151)

Explain Question : Identify correct statements about arthropods.
Concept : This question is based on Arthropoda.
Explanation : The correct answer is option 4. The correct statements are A,C,D .
(A) Arthropods have a chitinous exoskeleton - **correct**
(C) Arthropods are bilaterally symmetrical, triploblastic, and coelomate - **correct**
(D) Arthropods usually have an open circulatory system - **correct**
(B) They are mostly dioecious (separate sexes), not monoecious - **Incorrect**
Final answer : 4

152)

Explain Question: Identify the correct location of the mantle cavity in molluscs.
Concept: This question is based on Mollusca

Solution: The mantle cavity is the space between the visceral hump and the mantle, housing gills, excretory, and reproductive openings; so the correct match is “hump and mantle.”
Final answer: 2. Hump and mantle

153)

NCERT, Pg. # 39

154)

NCERT, Pg. # 39

155)

NCERT, Pg. # 40

156)

NCERT, Pg. # 40

157)

NCERT, Pg. # 45

158)

Explanation: The question asks which structure is absent in bacteria, i.e., not a component of a prokaryotic cell.
Concept: Prokaryotic cell structure (Bacteria)
Solution : Mitochondria → Not present in bacteria (they lack membrane-bound organelles). Instead, respiration occurs in the plasma membrane.
Final Answer: Mitochondria

159)

NCERT XI Pg.# 89-90

160)

NCERT XI Pg.# 87

161)

NCERT XI Pg.# 88

162)

NCERT XI Pg.# 88

163)

NCERT XI Pg.# 87-88

164)

The correct answer is : 1. a large central vacuole and rigid cell wall

Explanation: Plant cells have a large central vacuole for storage and a rigid cell wall for structure. Other options describe features not typical of plant cells.

165)

NCERT XI Pg.# 88

166) NCERT-XI, Pg.# 89

167)

NCERT XI Pg.# 87

168)

NCERT XI Pg.# 87-88

169) NCERT (XI) Pg. # 94 (8.5.2)

170)

NCERT - Pg. # 87

171)

NCERT-XI, Pg. # 93

172) NCERT-XI, Pg. # 94

173)

NCERT XI Pg.# 93

174)

NCERT-XI, Pg.# 93,94

175) Module Pg. # 170-172

176)

NCERT XI Pg.# 93

177)

NCERT-XI, Pg. # 93

178)

NCERT-XI, Pg.# 94

179) NCERT-XI, Pg.# 93,94

180) **Question Explanation :**
The question asks : Why does the cytoplasm of eukaryotic cells have so much compartmentalization.
Concept :
Eukaryotic Cells.
Solution :
The correct answer is 3. Due to the presence of membranous organelles
Explanation :
Eukaryotic cells have membrane – bound organelles (e.g., nucleus, ER, Golgi, mitochondria) These organelles create internal compartments, each specialized for specific functions (e.g. protein synthesis, energy production)
Final Answer :
Option (3)